

Assignment Problem by Branch and Bound

1 Problem Definition

Assignment Problem

There are n people and n jobs. Each person must receive exactly one job, each job must be assigned to exactly one person, and the total cost must be minimized.

For the example:

$$\text{People} = \{A, B, C, D\}, \quad \text{Jobs} = \{J_1, J_2, J_3, J_4\}.$$

The cost matrix is:

Person	J_1	J_2	J_3	J_4
A	9	2	7	8
B	6	4	3	7
C	5	8	1	8
D	7	6	9	4

Objective

Find a one-to-one assignment minimizing

$$\sum_{i=1}^n c_{i,\pi(i)}$$

where $\pi(i)$ is the job assigned to person i .

2 Lower Bound

At a partial assignment, some people and jobs are already fixed. The lower bound is:

$$LB = \text{fixed cost} + \sum \text{minimum available cost for each unassigned person.}$$

This is a lower bound because it relaxes the future assignment problem: each remaining person is allowed to choose the cheapest still-available job independently. The true completion cannot cost less than this optimistic estimate.

3 Root Bound

At the root, no job is fixed. Take the minimum entry from each row:

$$\min(A) = 2, \quad \min(B) = 3, \quad \min(C) = 1, \quad \min(D) = 4.$$

Therefore,

$$LB_{\text{root}} = 2 + 3 + 1 + 4 = 10.$$

Root Node

$$LB_{\text{root}} = 10.$$

4 Branch on Person A

Create four branches by assigning A to each job. Once a job is chosen, it is unavailable to the remaining people.

Branch	Lower-bound calculation	LB
$A \rightarrow J_1$	$9 + \min(4, 3, 7) + \min(8, 1, 8) + \min(6, 9, 4)$	17
$A \rightarrow J_2$	$2 + \min(6, 3, 7) + \min(5, 1, 8) + \min(7, 9, 4)$	10
$A \rightarrow J_3$	$7 + \min(6, 4, 7) + \min(5, 8, 8) + \min(7, 6, 4)$	20
$A \rightarrow J_4$	$8 + \min(6, 4, 3) + \min(5, 8, 1) + \min(7, 6, 9)$	18

The smallest live lower bound is:

$$A \rightarrow J_2, \quad LB = 10.$$

So this node is expanded next.

5 Branch on Person B

Now $A \rightarrow J_2$ is fixed. Job J_2 is unavailable, so B may take J_1, J_3 , or J_4 .

Branch	Lower-bound calculation	LB
$B \rightarrow J_1$	$2 + 6 + \min(1, 8) + \min(9, 4)$	13
$B \rightarrow J_3$	$2 + 3 + \min(5, 8) + \min(7, 4)$	14
$B \rightarrow J_4$	$2 + 7 + \min(5, 1) + \min(7, 9)$	17

The best live node is:

$$A \rightarrow J_2, \quad B \rightarrow J_1, \quad LB = 13.$$

6 Complete the Best Branch

With $A \rightarrow J_2$ and $B \rightarrow J_1$ fixed, the remaining jobs are J_3 and J_4 . The remaining people are C and D .

Completion	Cost
$C \rightarrow J_3, D \rightarrow J_4$	$2 + 6 + 1 + 4 = 13$
$C \rightarrow J_4, D \rightarrow J_3$	$2 + 6 + 8 + 9 = 25$

The first completion gives a complete feasible solution:

$$\text{bestCost} = 13.$$

Incumbent

$$A \rightarrow J_2, \quad B \rightarrow J_1, \quad C \rightarrow J_3, \quad D \rightarrow J_4$$

with total cost

13.

7 Pruning Remaining Nodes

For minimization, prune a live node when

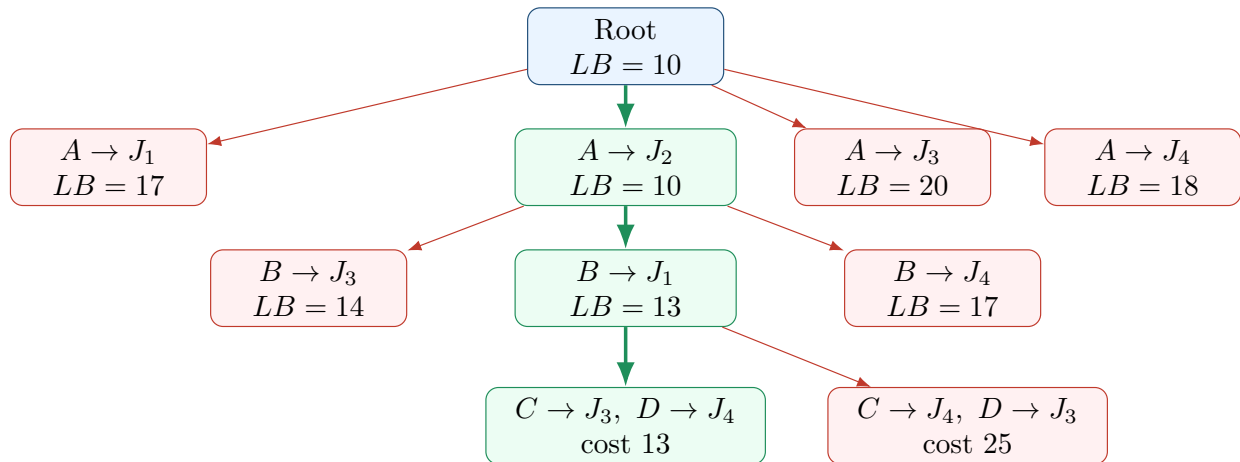
$$LB \geq \text{bestCost}.$$

Here $\text{bestCost} = 13$.

Remaining live node	LB	Decision
$A \rightarrow J_2, B \rightarrow J_3$	14	prune
$A \rightarrow J_2, B \rightarrow J_4$	17	prune
$A \rightarrow J_1$	17	prune
$A \rightarrow J_3$	20	prune
$A \rightarrow J_4$	18	prune

Every remaining live node has lower bound at least 13, so no branch can improve the incumbent.

8 State-Space Tree



9 Optimal Assignment

Person	Job	Cost
A	J_2	2
B	J_1	6
C	J_3	1
D	J_4	4

Minimum total cost = $2 + 6 + 1 + 4 = 13$.

10 Complexity

The worst-case number of possible assignments is the number of permutations of n jobs:

$$n!.$$

Therefore,

$$\boxed{\mathcal{O}(n!)}$$

is the worst-case time complexity for this Branch and Bound formulation.

Branch and Bound often expands far fewer nodes in practice, but the worst case remains factorial.

11 Final Review

Assignment Branch and Bound Essentials

$LB = \text{fixed cost} + \sum \text{row minima over available jobs.}$
For minimization, prune when $LB \geq \text{bestCost}$.
Best assignment: $A \rightarrow J_2, B \rightarrow J_1, C \rightarrow J_3, D \rightarrow J_4$.
Minimum cost = 13.